Prompt 1: How many pills are in this image? Answer with just a number.

Image ID	True Count	Variable Tested	Model	Model Answer	Correct?	Off by	Model's reasoning notes
1	7	baseline	ChatGPT	7	1	0	
2	3	count	ChatGPT	3	1	0	
3	5	count	ChatGPT	5	1	0	
4	10	count	ChatGPT	10	1	0	
5	15	count	ChatGPT	12	0	-3	
6	7	spacing	ChatGPT	8	0	1	
7	7	touching	ChatGPT	7	1	0	
8	7	overlapping	ChatGPT	6	0	-1	
9	7	background	ChatGPT	8	0	1	
10	7	background	ChatGPT	7	1	0	
11	7	lighting	ChatGPT	7	1	0	
12	7	shadows	ChatGPT	7	1	0	
13	7	color contrast	ChatGPT	8	0	1	
14	7	color uniformity	ChatGPT	8	0	1	
15	7	normal	ChatGPT	7	1	0	
1	7	baseline	Claude	7	1	0	
2	3	count	Claude	3	1	0	
3	5	count	Claude	5	1	0	
4	10	count	Claude	10	1	0	
5	15	count	Claude	15	1	0	
6	7	spacing	Claude	6	0	-1	
7	7	touching	Claude	6	0	-1	
8	7	overlapping	Claude	3	0	-4	
9	7	background	Claude	7	1	0	
10	7	background	Claude	7	1	0	
11	7	lighting	Claude	7	1	0	
12	7	shadows	Claude	7	1	0	
13	7	color contrast	Claude	7	1	0	
14	7	color uniformity	Claude	7	1	0	
15	7	normal	Claude	6	0	-1	
1	7	baseline	Gemini	7	1	0	
2	3	count	Gemini	3	1	0	
3	5	count	Gemini	5	1	0	
4	10	count	Gemini	10	1	0	
5	15	count	Gemini	15	1	0	
6	7	spacing	Gemini	7	1	0	
7	7	touching	Gemini	7	1	0	
8	7	overlapping	Gemini	6	0	-1	
9	7	background	Gemini	7	1	0	
10	7	background	Gemini	7	1	0	
11	7	lighting	Gemini	7	1	0	
12	7	shadows	Gemini	7	1	0	
13	7	color contrast	Gemini	7	1	0	
14	7	color uniformity	Gemini	7	1	0	
15	7	normal	Gemini	7	1	0	

Prompt 2: I need you to count every pill in this image carefully. Go through the image systematically, count each visible pill, then give me the final total.

Image ID	True Count	Variable Tested	Model	Model Answer	Correct?	Off by	Model's reasoning notes
1	7	baseline	ChatGPT	8	0	1	
2	3	count	ChatGPT	3	1	0	
3	5	count	ChatGPT	5	1	0	
4	10	count	ChatGPT	9	0	-1	
5	15	count	ChatGPT	14	0	-1	
6	7	spacing	ChatGPT	7	1	0	
7	7	touching	ChatGPT	6	0	-1	
8	7	overlapping	ChatGPT	6	0	-1	
9	7	background	ChatGPT	7	1	0	
10	7	background	ChatGPT	7	1	0	
11	7	lighting	ChatGPT	7	1	0	
12	7	shadows	ChatGPT	8	0	1	
13	7	color contrast	ChatGPT	8	0	1	
14	7	color uniformity	ChatGPT	7	1	0	
15	7	normal	ChatGPT	7	1	0	
1	7	baseline	Claude	7	1	0	
2	3	count	Claude	3	1	0	
3	5	count	Claude	5	1	0	
4	10	count	Claude	10	1	0	
5	15	count	Claude	15	1	0	
6	7	spacing	Claude	6	0	-1	
7	7	touching	Claude	4	0	-3	
8	7	overlapping	Claude	3	0	-4	
9	7	background	Claude	7	1	0	
10	7	background	Claude	7	1	0	
11	7	lighting	Claude	7	1	0	
12	7	shadows	Claude	7	1	0	
13	7	color contrast	Claude	7	1	0	
14	7	color uniformity	Claude	7	1	0	
15	7	normal	Claude	6	0	-1	
1	7	baseline	Gemini	7	1	0	
2	3	count	Gemini	3	1	0	
3	5	count	Gemini	5	1	0	
4	10	count	Gemini	10	1	0	
5	15	count	Gemini	15	1	0	
6	7	spacing	Gemini	7	1	0	
7	7	touching	Gemini	7	0	0	
8	7	overlapping	Gemini	7	0	0	
9	7	background	Gemini	7	1	0	
10	7	background	Gemini	7	1	0	
11	7	lighting	Gemini	7	1	0	
12	7	shadows	Gemini	7	1	0	
13	7	color contrast	Gemini	7	1	0	
14	7	color uniformity	Gemini	7	1	0	
15	7	normal	Gemini	7	1	0	

Prompt 3: How many pills are in this image? After you give your answer, double-check by counting again and confirm or correct your number.

Image ID	True Count	Variable Tested	Model	Model Answer	Correct?	Off by	Model's reasoning notes
1	7	baseline	ChatGPT				
2	3	count	ChatGPT				
3	5	count	ChatGPT				
4	10	count	ChatGPT				
5	15	count	ChatGPT				
6	7	spacing	ChatGPT				
7	7	touching	ChatGPT				
8	7	overlapping	ChatGPT				
9	7	background	ChatGPT				
10	7	background	ChatGPT				
11	7	lighting	ChatGPT				
12	7	shadows	ChatGPT				
13	7	color contrast	ChatGPT				
14	7	color uniformity	ChatGPT				
15	7	normal	ChatGPT				
1	7	baseline	Claude				
2	3	count	Claude				
3	5	count	Claude				
4	10	count	Claude				
5	15	count	Claude				
6	7	spacing	Claude				
7	7	touching	Claude				
8	7	overlapping	Claude				
9	7	background	Claude				
10	7	background	Claude				
11	7	lighting	Claude				
12	7	shadows	Claude				
13	7	color contrast	Claude				
14	7	color uniformity	Claude				
15	7	normal	Claude				
1	7	baseline	Gemini				
2	3	count	Gemini				
3	5	count	Gemini				
4	10	count	Gemini				
5	15	count	Gemini				
6	7	spacing	Gemini				
7	7	touching	Gemini				
8	7	overlapping	Gemini				
9	7	background	Gemini				
10	7	background	Gemini				
11	7	lighting	Gemini				
12	7	shadows	Gemini				
13	7	color contrast	Gemini				
14	7	color uniformity	Gemini				
15	7	normal	Gemini				

Low counts are easy for all models
High counts break some models (especially ChatGPT)
SPACING / OVERLAP is the BIGGEST failure mode
Background + lighting matter LESS than expected
Color contrast matters (especially for ChatGPT)

“Double-checking” ≠ actually seeing better
Even when models are explicitly asked to double-check their answer, they do not reliably correct mistakes. Instead, they often repeat the same incorrect reasoning in a more structured way.

Counting might improve when the task is decomposed into subgroups - colors gmeini

Overlap/touching = biggest failure
High counts break some models (ChatGPT)
Prompting does not reliably fix errors
Models explain mistakes confidently instead of correcting them

Scan the image from left to right and top to bottom. Count each pill as you encounter it and keep a running total. Then give the final count.
Group the pills by color or type, count how many are in each group, then add them together to get the total number of pills.
First count the pills once.
Then independently count them again in a different order.
If the counts differ, explain why and give your best estimate.
Boxing

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P1	Naive baseline	How many pills are in this image?	Control - no guidance at all
P2	Careful one-by-one	Look at this image carefully. Count each individual pill one by one before	Whether slowing down helps (basic chain-of-
P3	Color-first decomposition	First, identify each distinct pill color you see. Then count how many pills there are of each color. Finally, add those subtotals to give the total pill	Structured decomposition by color grouping, maps to pill verification use case
P4	Bounding box detection	Detect each pill in this image. For every pill, provide its approximate bounding box location as [top, left, bottom, right] and its color. Return the	Uses Gemini's trained spatial detection to force commitment to specific pill regions before
P5	Self-verification recount	Count all the pills in this image. Then recount them using a different method. If your two counts disagree, explain the discrepancy and give your best final	Metacognition - can the model catch its own errors?
P6	Row-by-row structured counting	Divide this image into a top half and bottom half. Count the pills in the top half first, then count the pills in the bottom half. Add the two numbers for the	Spatial partitioning reduces estimation errors (supported by research)
P7	Medication safety persona + uncertainty	You are a medication adherence assistant helping an elderly patient verify their pills before they take them. Count every visible pill, identify each pill's	Role-prompting + stakes framing + uncertainty flagging (addresses lit review gap)
P8	Describe scene first, then count	First, describe everything you see in this image in detail, including the surface, the arrangement of the pills, their colors, and any pills that might be touching or overlapping. After your description, count the total number of	Forces full visual processing before counting, prevents jumping to estimates
P9	Uncertainty-aware with confidence	Count the pills in this image. For your answer, provide: (1) your count, (2) a confidence level (high/medium/low), and (3) explain what made counting	Self-assessment of confidence - key for safety-critical use
P10	Numbered mental labeling	Imagine you are placing a small numbered sticker on each pill in this image, starting with 1. For each pill, state its number, its color, and where it is in the	Inspired by Set-of-Mark (SoM) research - assigning individual IDs improves counting

Image ID	True Count	Pill Colors	Difficulty	Prompt ID	Prompt Type	Model Answer	Correct?	Error (±)	Confidence (if given)	Color Accuracy	Notes
IMG-01	7	green, 2 white, 2 brow	Easy	P1	Naive baseline	7	Yes	0			
IMG-01	7	green, 2 white, 2 brow	Easy	P2	Careful counting						
IMG-01	7	green, 2 white, 2 brow	Easy	P3	Color-first decomposition						
IMG-01	7	green, 2 white, 2 brow	Easy	P4	Spatial scanning						
IMG-01	7	green, 2 white, 2 brow	Easy	P5	Self-verification						
IMG-01	7	green, 2 white, 2 brow	Easy	P6	Location enumeration						
IMG-01	7	green, 2 white, 2 brow	Easy	P7	Medication safety persona						
IMG-01	7	green, 2 white, 2 brow	Easy	P8	Structured JSON output						
IMG-01	7	green, 2 white, 2 brow	Easy	P9	Uncertainty-aware						
IMG-01	7	green, 2 white, 2 brow	Easy	P10	Multi-pass physical reasoning						
IMG-02	7	green, 2 white, 2 brow	Medium	P1	Naive baseline						
IMG-02	7	green, 2 white, 2 brow	Medium	P2	Careful counting						
IMG-02	7	green, 2 white, 2 brow	Medium	P3	Color-first decomposition						
IMG-02	7	green, 2 white, 2 brow	Medium	P4	Spatial scanning						
IMG-02	7	green, 2 white, 2 brow	Medium	P5	Self-verification						
IMG-02	7	green, 2 white, 2 brow	Medium	P6	Location enumeration						
IMG-02	7	green, 2 white, 2 brow	Medium	P7	Medication safety persona						
IMG-02	7	green, 2 white, 2 brow	Medium	P8	Structured JSON output						
IMG-02	7	green, 2 white, 2 brow	Medium	P9	Uncertainty-aware						
IMG-02	7	green, 2 white, 2 brow	Medium	P10	Multi-pass physical reasoning						
IMG-03	7	green, 2 white, 2 brow	Hard	P1	Naive baseline						
IMG-03	7	green, 2 white, 2 brow	Hard	P2	Careful counting						
IMG-03	7	green, 2 white, 2 brow	Hard	P3	Color-first decomposition						
IMG-03	7	green, 2 white, 2 brow	Hard	P4	Spatial scanning						
IMG-03	7	green, 2 white, 2 brow	Hard	P5	Self-verification						
IMG-03	7	green, 2 white, 2 brow	Hard	P6	Location enumeration						
IMG-03	7	green, 2 white, 2 brow	Hard	P7	Medication safety persona						
IMG-03	7	green, 2 white, 2 brow	Hard	P8	Structured JSON output						
IMG-03	7	green, 2 white, 2 brow	Hard	P9	Uncertainty-aware						
IMG-03	7	green, 2 white, 2 brow	Hard	P10	Multi-pass physical reasoning						
IMG-04	15	green, 5 white, 5 brow	Hard	P1	Naive baseline						
IMG-04	15	green, 5 white, 5 brow	Hard	P2	Careful counting						
IMG-04	15	green, 5 white, 5 brow	Hard	P3	Color-first decomposition						
IMG-04	15	green, 5 white, 5 brow	Hard	P4	Spatial scanning						
IMG-04	15	green, 5 white, 5 brow	Hard	P5	Self-verification						
IMG-04	15	green, 5 white, 5 brow	Hard	P6	Location enumeration						
IMG-04	15	green, 5 white, 5 brow	Hard	P7	Medication safety persona						
IMG-04	15	green, 5 white, 5 brow	Hard	P8	Structured JSON output						
IMG-04	15	green, 5 white, 5 brow	Hard	P9	Uncertainty-aware						
IMG-04	15	green, 5 white, 5 brow	Hard	P10	Multi-pass physical reasoning						

Prompt ID	Prompt Type	# Correct (of 4)	Accuracy %	Mean Abs	Overcounts	Undercounts
P1	Naive baseline	1	25%	0.0	3	0
P2	Careful counting	0	0%	0.0	4	0
P3	Color-first decomposition	0	0%	0.0	4	0
P4	Spatial scanning	0	0%	0.0	4	0
P5	Self-verification	0	0%	0.0	4	0
P6	Location enumeration	0	0%	0.0	4	0
P7	Medication safety persona	0	0%	0.0	4	0
P8	Structured JSON output	0	0%	0.0	4	0
P9	Uncertainty-aware	0	0%	0.0	4	0
P10	Multi-pass physical reasoning	0	0%	0.0	4	0

Image ID	True Count	Pill Colors	Condition	Description	File Name
IMG-01	7	3 green, 2 white, 2	Baseline/normal	7 pills, clean background, good lighting, not touching	(fill in)
IMG-02	7	3 green, 2 white, 2	Touching	7 pills, some pills touching each other	(fill in)
IMG-03	7	3 green, 2 white, 2	Overlapping	7 pills, some pills overlapping — hardest condition	(fill in)
IMG-04	15	5 green, 5 white, 5	Normal	15 pills, normal conditions, high count challenge	(fill in)